



Original article

Vitamin and mineral supplements: Are they associated with fatigue?

Sisi Xie^a, Pedro Marques-Vidal^a, Vanessa Kraege^{a, b, c, *}^a Department of Medicine, Internal Medicine, Lausanne University Hospital (CHUV) and University of Lausanne, Lausanne, Switzerland^b Medical Directorate, Lausanne University Hospital (CHUV) and University of Lausanne, Lausanne, Switzerland^c Innovation and Research Directorate, Lausanne University Hospital (CHUV) and University of Lausanne, Lausanne, Switzerland

ARTICLE INFO

Article history:

Received 2 June 2024

Accepted 29 October 2024

Keywords:

Vitamins

Minerals

Dietary supplements

Fatigue

Epidemiology

SUMMARY

Background and aims: Vitamin/mineral supplements (VMS) and vitamin/mineral and/or dietary supplements (VMDS) are commonly consumed, based on the belief of their invigorating properties. Still, the association of VMS/VMDS with fatigue in the general population remains sparse and conclusions are mixed. This study aimed to understand the association between VMS/VMDS and fatigue.

Methods: Two cross-sectional studies (2014–2017 and 2018–2021) conducted on a population-based cohort. Participants were divided into VMS/VMDS users and non-users. Fatigue levels were assessed using the Fatigue Severity Scale (FSS) and the 14-item version of the Chalder Fatigue Scale (CFS).

Results: Overall, 2084 (50.8 % female, 61.7 ± 9.8 years) and 1728 (51.0 % female, 64.1 ± 9.3 years) were included in the first and second surveys, respectively. In the first survey, after multivariable analysis, no association was found between VMS use and FSS: adjusted mean ± sem 3.04 ± 0.13 vs. 2.82 ± 0.03 for users and non-users, respectively, $p = 0.083$, while users scored higher in the CFS: 6.17 ± 0.22 vs. 5.72 ± 0.06 for users and non-users, respectively, $p = 0.048$. In the second survey, no association was found between VMS use and FSS (2.83 ± 0.19 vs. 2.84 ± 0.03, $p = 0.952$) and CFS (5.38 ± 0.33 vs. 5.61 ± 0.06, $p = 0.504$). In the first survey, VMDS users scored higher in both FSS (3.03 ± 0.06 vs. 2.76 ± 0.04, $p < 0.001$) and CFS (6.19 ± 0.11 vs. 5.58 ± 0.06, $p < 0.001$) while no differences were found in the second survey in both FSS (2.93 ± 0.07 vs. 2.81 ± 0.04, $p = 0.161$) and CFS (5.78 ± 0.12 vs. 5.54 ± 0.07, $p = 0.088$).

Conclusion: In this population-based cross-sectional study, the authors found no consistent association between VMS consumption and fatigue. Participants taking VMDS tended to present with higher fatigue scores.

© 2024 The Author(s). Published by Elsevier Ltd on behalf of European Society for Clinical Nutrition and Metabolism. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Fatigue is a common symptom in primary care, usually characterized by low energy, mental exhaustion, and poor muscle endurance. It can have a negative impact on work, family, and society [1]. In healthy people, fatigue is a physiological response to prolonged strenuous physical or mental exertion. It is short-lived and predictable, decreases with rest, and usually does not interfere with daily activities [2]. However, fatigue in sick people has different characteristics and is described as an overwhelming

tiredness, lack of energy, and feeling of exhaustion that cannot be compared with physiological fatigue after physical or mental exertion and cannot be recovered by sleep [3]. For ill people to improve fatigue symptoms, the underlying disease should be treated actively. However, in healthy individuals, the underlying reason for symptoms is often unknown, and treatment options are scarce; thus, nutritional supplements have become a common response, despite lack of scientific evidence [4].

Vitamin/mineral supplements (VMS) are common nutritional supplements that provide essential elements for humans. According to the 2011–2014 National Health and Nutrition Examination Survey, half (52 %) of U.S. adults reported using at least one dietary supplement, and nearly one-third (31 %) reported using a VMS [5]. On the one hand, vitamins and minerals are involved in important physiological functions such as energy-producing metabolism, DNA synthesis, oxygen transport, and neuronal function. Their

* Corresponding author. Medical directorate and Innovation and Clinical Research directorate, Centre Hospitalier Universitaire Vaudois-CHUV, Lausanne University Hospital Bugnon 21, Bureau 21-06-247, 1011 Lausanne, Switzerland.

E-mail addresses: Sisi.Xie@unil.ch (S. Xie), Pedro-Manuel.Marques-Vidal@chuv.ch (P. Marques-Vidal), vanessa.kraege@chuv.ch (V. Kraege).

deficiency can directly cause diseases in the body, leading to fatigue symptoms. On the other hand, vitamin and mineral deficiencies can cause mental fatigue by translating into effects on cognitive and mental processes [6].

Therefore, fatigue symptoms caused by insufficient vitamins and minerals can usually be treated through active supplementation. However, research on the potential association of VMS with fatigue in the general population remains sparse and conclusions are mixed. Hence, given the lack of scientific evidence, this study focused on the association between VMS and/or dietary (VMDs) supplements and fatigue in the general population.

2. Participants and methods

2.1. Population and study design

This study used data from CoLausPsyCoLaus (www.colauspsycolaus.ch). CoLausPsyCoLaus study is a population-based study initiated in 2003 with 6733 middle-aged participants from Lausanne, Switzerland, to investigate the epidemiology and genetic determinants of cardiovascular risk factors [7]. Recruitment began in 2003 and ended in 2006. The first follow-up was performed between 2009 and 2012, the second between 2014 and 2017 and the third between 2018 and 2021. In this study, the authors used the 2nd and 3rd follow-up data for cross-sectional analysis, as they contained the necessary variables.

2.2. Vitamin/mineral/dietary supplements consumption

Participants were asked to report all prescribed and over-the-counter medications and supplements taken regularly over the last six months. Vitamin and mineral supplements were defined according to the Swiss compendium (compendium.ch/home/fr, assessed June 2017). When the supplements were not listed in the Swiss compendium, further searches were conducted on the internet. Due to wide differences in the composition of Swiss VMS [8] and to inaccurate reporting (i.e., reporting “multivitamins from producer X” that manufactures six different types of multivitamins), it was not possible to assess the amounts of vitamins and minerals consumed by participants. Dietary supplements were defined as any other supplement that could not be considered as a VMS, such as plant extracts not considered as phytotherapy by the Swiss compendium, cod liver oil, shark cartilage or amino acids.

2.3. Assessment of fatigue

Fatigue levels were assessed using the Fatigue Severity Scale (FSS) and the 14-item version of the Chalder Fatigue Scale (CFS), which were filled by the participants.

Fatigue over the previous week was assessed by the 9 items FSS [9]. This questionnaire has been validated for a general healthy population in the Swiss setting [10] and has a high test-retest reliability [11]. The questionnaire is composed of nine questions; responses are graded using a Likert scale from 1 to 7, where 1 indicates strong disagreement and 7 strong agreement. The final score is the mean value of the 9 responses, and a score ≥ 4 is considered as severe fatigue. Fatigue was also assessed using the 14-item CFS [12]. In this study, a binary coding (presence/absence) to the items was applied, as it has been shown that near-maximal scoring of the 14 items constituting the Chalder fatigue scale supports the validity of a two-point scoring rather than the four-point Likert scoring [13].

Cronbach α coefficient was used to evaluate internal consistency and reliability, and the results indicated that the internal reliability of both FSS and CFS was very good (Cronbach's alpha were 0.918

and 0.866 respectively). While the CFS has good internal consistency [14,15] and convergent validity [16], it also presents ceiling effects [17], as most fatigued patients score the highest value.

2.4. Relevant covariates

This study selected potential confounding factors based on the literature on the relationship between VMS and Fatigue. Authors selected age (years), sex (male/female), education, marital status, weekly alcohol consumption (units), smoking, hypertension (yes/no), diabetes (yes/no), body mass index (BMI) categories, total energy intake (kcal/day), and Alternate Healthy Eating Index (AHEI).

Education was categorized into high (university), middle (high school) and low (apprenticeship + mandatory). Marital status was defined as living alone (single, divorced, widowed) or living with a partner. Usual alcohol consumption during the week was self-reported as number of units (glasses of wine, bottles or cans of beer, and shots of spirits) per week. Smoking was self-reported and categorized as never, former (irrespective of the time since quitting smoking) and current.

Body weight and height were measured with participants barefoot and in light indoor clothes. Body weight was measured in kilograms to the nearest 100 g using a Seca® scale (Hamburg, Germany). Height was measured to the nearest 5 mm using a Seca® (Hamburg, Germany) height gauge. BMI was calculated and categorized as normal ($<25 \text{ kg/m}^2$), overweight ≥ 25 and $<30 \text{ kg/m}^2$ and obese $\geq 30 \text{ kg/m}^2$.

Blood pressure (BP) was measured using an Omron® HEM-907 automated oscillometric sphygmomanometer after at least a 10-min rest in a seated position, and the average of the last two measurements was used. Hypertension was defined by a SBP $\geq 140 \text{ mm Hg}$ or a DBP $\geq 90 \text{ mm Hg}$ or presence of antihypertensive drug treatment.

Glucose was assessed by glucose dehydrogenase. Diabetes mellitus (DM) was defined as fasting plasma glucose $\geq 7.0 \text{ mmol/L}$ and/or presence of oral hypoglycaemic or insulin treatment.

The AHEI was adapted from McCullough et al. [18]. In our study, the amount of trans fat could not be assessed, and in a first analysis, the authors considered all participants taking multivitamins as taking them for a duration ≥ 5 years. Thus, the modified AHEI score ranged between 2.5 and 77.5 instead of 2.5 and 87.5 for the original AHEI score. As it was not possible to assess whether participants were taking multivitamins for the last 5 years, a second AHEI was computed after removing the multivitamin component. This vitamin-free AHEI ranged between 2.5 and 70.0. The Spearman correlation between the vitamin-free and the modified AHEI was 0.984. For both AHEIs, higher values represented a healthier diet.

2.5. Exclusion criteria

This study excluded participants with missing data for 1) fatigue; 2) covariates.

2.6. Ethical statement

The institutional Ethics Committee of the University of Lausanne, which afterwards became the Ethics Commission of Canton Vaud (www.cer-vd.ch) approved the CoLaus-PsyCoLaus study (project number PB_2018–00038, reference 239/09). All participants gave their signed informed consent before entering the study.

2.7. Statistical analysis

Statistical analyses were conducted using Stata v.18 (Stata Corp, College Station, TX, USA). Descriptive results were expressed as

number of participants (percentage) for categorical variables and as average $\pm$ standard deviation for continuous variables. Between-group comparisons were conducted using chi-square for categorical variables and student's t-test for continuous variables. Multivariable analyses were conducted using logistic regression for categorical variables, and results were expressed as odds ratio and (95 % confidence interval). Multivariable analyses were conducted using analysis of variance (ANOVA) for quantitative variables and results were expressed as adjusted mean $\pm$ standard error. Multivariable analyses were adjusted for the covariates defined previously. Statistical significance was considered for a two-sided test with $p < 0.05$.

3. Results

3.1. Selection of participants

Of the initial 4881 participants in the 2nd follow-up, 2084 (42.7 %) were included in the analyses. Of the initial 3482 in the 3rd follow-up, 1728 (49.6 %) were included in the analyses. The reasons for exclusion are indicated in Figure 1 and the comparison between included and excluded participants is provided in Supplementary Table 1. In both follow-ups, excluded participants were older, more frequently female, with lower education level, living alone, abstainers from alcohol, hypertensive or diabetic, and had lower total energy intake. Excluded participants scored higher than included participants regarding FSS and CFS at the 2nd follow-up, but there was no difference upon 3rd follow-up.

3.2. Association between VMS and VMDS consumption and fatigue

Among participants enrolled in the study at the 2nd follow-up, 129 (6.2 %) consumed VMS and 563 (27.0 %) consumed VMDS. The corresponding numbers for the 3rd follow-up were 55 (3.2 %) and 262 (23.4 %). Tables 1 and 2 summarize the characteristics of participants according to VMS and VMDS consumption in the 2nd and 3rd follow-ups, respectively. In the 2nd follow-up, compared with participants who did not consume VMS, participants who consumed VMS were more frequently female, lived alone, and had a higher AHEI (Table 1). Compared with participants who did not consume VMDS, participants who consumed VMDS were older, more frequently female, highly educated, non-obese, lived alone and had a higher AHEI (Table 1). In the 3rd follow-up, compared

with participants who did not consume VMS, participants who consumed VMS presented less frequently with hypertension, and had a higher AHEI (Table 2). Compared with participants who did not consume VMDS, participants who consumed VMDS were older, more frequently female, non-obese, less often current smokers, lived alone, and had a higher AHEI (Table 2).

For the 2nd follow-up, a summary of the association between VMS consumption and fatigue is shown in Table 3. In bivariate and multivariable analyses, there was no association between VMS consumption and FSS and prevalence of clinical fatigue. However, participants consuming VMS had higher scores for CFS (Table 3). A summary of the association between VMDS consumption and fatigue is shown in Table 4. In bivariate and multivariable analyses, participants consuming VMDS at second follow-up had higher scores for FSS and CFS, and more frequently had clinical fatigue (Table 4). To examine the robustness of the results, when the authors adjusted the multivariable analysis using the vitamin-free AHEI, the results were completely consistent for VMDS, whereas for VMS, the association between VMS consumption and higher CFS scores disappeared (Supplementary Table 2 and Supplementary Table 3).

For the 3rd follow-up, a summary of the association between VMS consumption and fatigue is shown in Table 3. In bivariate and multivariable analyses, there was no association between VMS consumption upon 3rd follow-up and FSS, CFS or prevalence of clinical fatigue (Table 4). A summary of the association between VMDS consumption upon 3rd follow-up and fatigue is shown in Table 4. In bivariate and multivariable analyses, there was no association between VMDS consumption and FSS. In bivariate analyses, participants consuming VMDS had higher scores for CFS, but the association was not significant in multivariable analysis. However, in bivariate and multivariable analyses, participants consuming VMDS more frequently had clinical fatigue (Table 4). To examine the robustness of the results, when the authors adjusted the multivariable analysis using the vitamin-free AHEI, the results were completely consistent (Supplementary Tables 2 and 3).

4. Discussion

In this study, the authors found no consistent association between VMS consumption and fatigue after adjusting for potential confounders. However, participants taking VMDS tended to present with higher fatigue scores.

4.1. Association between VMS consumption and fatigue

Research on the potential association between VMS, dietary supplements, and fatigue in the general population remains sparse and conclusions are mixed. A study conducted by Suzuki et al. showed that supplementing 16 young athletes with high-dose thiamine (vitamin B₁) for 3 days after cycling exercise increased blood thiamine levels and reduced the number of fatigue complaints [19]. A double-blind randomized controlled trial showed that intravenous vitamin C reduced fatigue within 2 h and the effects lasted for a day [20]. In another study that included 290 women, 1g intravenous iron reduced fatigue in 65.3 % of the intervention group, compared to 52.7 % in the placebo group, a significant difference [21]. Among healthy elderly people with low physical activity levels, vitamin D and calcium might prevent muscle fatigue by regulation of the biosynthesis of creatine kinase, lactic acid dehydrogenase, and troponin I [22]. In addition, vitamin D treatment significantly improved fatigue in healthy people deficient in vitamin D [23,24]. Finally, vitamin E supplementation also

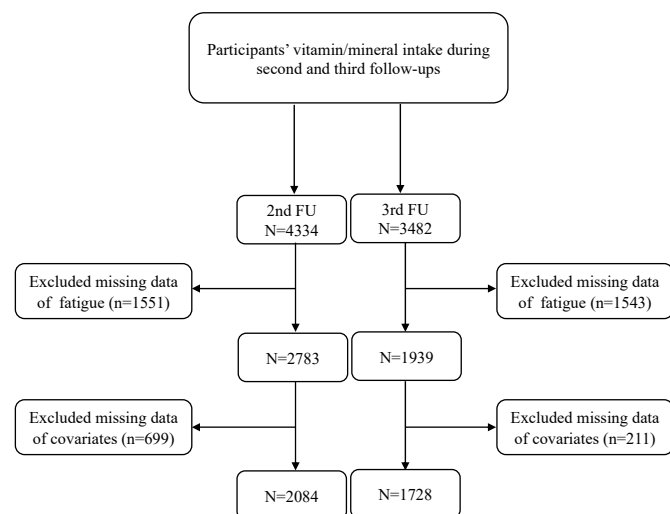


Fig. 1. Selection of participants from the second and third follow-ups. CoLaus study, Lausanne, Switzerland.

Table 1

Characteristics of the participants, according to VMS and VMDS consumption, second follow-up of the CoLaus|PsyColaus study, Lausanne, Switzerland.

	VMS Non-users	VMS Users	P-value	VMDS Non-users	VMDS Users	P-value
Sample size (%)	1955 (93.8)	129 (6.2)		1521 (73.0)	563 (27.0)	
Age (years)	61.6 ± 9.8	62.8 ± 9.7	0.197	61.2 ± 9.6	63.1 ± 10	<0.001
Women (%)	974 (49.8)	84 (65.1)	0.001	689 (45.3)	369 (65.5)	<0.001
Educational level (%)			0.180			0.009
Low	905 (46.3)	52 (40.3)		728 (47.9)	229 (40.7)	
Middle	581 (29.7)	37 (28.7)		442 (29.1)	176 (31.3)	
High	469 (24.0)	40 (31.0)		351 (23.1)	158 (28.1)	
Marital status, %			<0.001			<0.001
Living alone	615 (31.5)	65 (50.4)		449 (29.5)	231 (41.0)	
Living in couple	1340 (68.5)	64 (49.6)		1072 (70.5)	332 (59.0)	
Alcohol consumption, %			0.791			0.023
None	407 (20.8)	31 (24)		297 (19.5)	141 (25.1)	
1–13/week	1255 (64.2)	79 (61.2)		990 (65.1)	344 (61.1)	
14–27/week	245 (12.5)	15 (11.6)		199 (13.1)	61 (10.8)	
28+/week	48 (2.5)	4 (3.1)		35 (2.3)	17 (3.0)	
Smoking categories (%)			0.827			0.105
Never	795 (40.7)	53 (41.1)		636 (41.8)	212 (37.6)	
Former	802 (41)	50 (38.8)		601 (39.5)	251 (44.6)	
Current	358 (18.3)	26 (20.2)		284 (18.7)	100 (17.8)	
BMI categories (%)			0.326			0.002
Normal	840 (43)	54 (41.9)		620 (40.8)	274 (48.7)	
Overweight	789 (40.4)	59 (45.7)		632 (41.5)	216 (38.3)	
Obese	326 (16.7)	16 (12.4)		269 (17.7)	73 (13.0)	
Hypertension (%)	862 (44.1)	51 (39.5)	0.312	673 (44.3)	240 (42.6)	0.508
Diabetes (%)	166 (8.5)	7 (5.4)	0.222	128 (8.4)	45 (8.0)	0.756
Total energy intake (kcal/d)	1735 ± 631	1688 ± 570	0.412	1733 ± 637	1729.7 ± 601.6	0.915
AHEI	32 ± 10	35.8 ± 10.3	<0.001	31.3 ± 9.8	34.7 ± 10.3	<0.001

Results are expressed as number of participants (column percentage) for categorical variables and as average ± standard deviation for continuous variables. Between-group comparisons performed using chi-square for categorical variables and student's t-test for continuous variables. Abbreviations: VMS: Vitamin/mineral supplements; VMDS: Vitamin/mineral and/or dietary supplements; AHEI: Alternate Healthy Eating Index.

Table 2

Characteristics of the participants, according to VMS and VMDS consumption, third follow-up of the CoLaus|PsyColaus study, Lausanne, Switzerland.

	VMS Non-users	VMS Users	P-value	VMDS Non-users	VMDS Users	P-value
Sample size (%)	1673 (96.8)	55 (3.2)		858 (76.6)	262 (23.4)	
Age (years)	64.0 ± 9.3	64.9 ± 9.0	0.486	63.6 ± 9.2	65.6 ± 9.5	<0.001
Women (%)	849 (50.8)	33 (60.0)	0.177	613 (46.3)	269 (66.6)	<0.001
Educational level (%)			0.195			0.236
Low	766 (45.8)	20 (36.4)		617 (46.6)	169 (41.8)	
Middle	498 (29.8)	16 (29.1)		386 (29.2)	128 (31.7)	
High	408 (24.4)	19 (34.6)		320 (24.2)	107 (26.5)	
Marital status, %			0.520			0.001
Living alone	748 (44.7)	27 (49.1)		564 (42.6)	211 (52.2)	
Living in couple	925 (55.3)	28 (50.9)		760 (57.4)	193 (47.8)	
Alcohol consumption, %			0.053			0.133
None	416 (24.9)	7 (12.7)		323 (24.4)	100 (24.8)	
1–13/week	998 (59.6)	43 (78.2)		784 (59.2)	257 (63.6)	
14–27/week	209 (12.5)	4 (7.3)		175 (13.2)	38 (9.4)	
28+/week	50 (3.0)	1 (1.8)		42 (3.2)	9 (2.2)	
Smoking categories (%)			0.191			0.015
Never	723 (43.2)	20 (36.4)		581 (43.9)	162 (40.1)	
Former	682 (40.8)	29 (52.7)		521 (39.3)	190 (47.0)	
Current	268 (16.0)	6 (10.9)		222 (16.8)	52 (12.9)	
BMI categories (%)			0.074			0.012
Normal	717 (42.8)	22 (40.0)		544 (41.1)	195 (48.3)	
Overweight	637 (38.1)	28 (50.9)		515 (38.9)	150 (37.1)	
Obese	319 (19.1)	5 (9.1)		265 (20.0)	59 (14.6)	
Hypertension (%)	826 (49.4)	17 (30.9)	0.007	663 (50.1)	180 (44.6)	0.052
Diabetes (%)	144 (8.6)	4 (7.3)	0.728	115 (8.7)	33 (8.2)	0.745
Total energy intake (kcal/d)	1634 ± 656	1717 ± 5545	0.358	1641 ± 672.6	1623.3 ± 586	0.633
AHEI	31.3 ± 9.9	38.7 ± 8.8	<0.001	31 ± 9.9	33.4 ± 10	<0.001

Results are expressed as number of participants (column percentage) for categorical variables and as average ± standard deviation for continuous variables. Between-group comparisons were performed using chi-square for categorical variables and student's t-test for continuous variables. Abbreviations: VMS: Vitamin/mineral supplements; VMDS: Vitamin/mineral and/or dietary supplements; AHEI: Alternate Healthy Eating Index.

improved subjective status and antioxidant capacity in healthy people with daily fatigue [25].

Regarding multivitamins or minerals, research by Lee et al. found that supplementing vitamin B complex for 28 consecutive days improved the exercise endurance performance of non-athletes

and reduced fatigue-related biochemical metabolites [26]. A placebo-controlled, double-blind, randomized, parallel-group trial by Haskell et al. showed that participants in the vitamin/mineral group exhibited an attenuation of the negative effects of extended task completion on mood/fatigue [27]. A prospective study by

Table 3

Bivariate and multivariable association VMS consumption and fatigue, CoLaus|PsyColaus study, Lausanne, Switzerland.

	Bivariate			Multivariable		
	VMS Non-users	VMS Users	P-value	VMS Non-users	VMS Users	P-value
2nd FU						
Fatigue score	2.82 ± 0.03	3.06 ± 0.13	0.064	2.82 ± 0.03	3.04 ± 0.13	0.083
Chalder score	5.70 ± 0.06	6.33 ± 0.23	0.008	5.72 ± 0.06	6.17 ± 0.22	0.048
Clinical fatigue (%)	402 (20.6)	32 (24.8)	0.250	1 (ref.)	1.33 (0.87–2.05)	0.191
3rd FU						
Fatigue score	2.84 ± 0.03	2.83 ± 0.19	0.948	2.84 ± 0.03	2.83 ± 0.19	0.952
Chalder score	5.61 ± 0.06	5.49 ± 0.33	0.732	5.61 ± 0.06	5.38 ± 0.33	0.504
Clinical fatigue (%)	335 (20.0)	12 (21.8)	0.744	1 (ref.)	1.30 (0.67–2.55)	0.440

Results are expressed as average ± standard deviation or number of participants (column percentage) for bivariate analysis and as multivariable-adjusted average ± standard error or odds ratio and 95 % confidence interval. Between group comparison using ANOVA or t-test for bivariate analyses and ANOVA or logistic regression for multivariable analyses, adjusting for age, sex, BMI categories (normal, overweight, obese), education (low/medium/high), marital status (alone, in couple), smoking (never, former, current), alcohol consumption (none, 1–13, 14–27 and 28+ per week), hypertension (yes, no), diabetes (yes, no), total energy intake (continuous), Alternate Healthy Eating Index (continuous). Abbreviations: VMS: Vitamin/mineral supplements; 2nd FU: second follow-up; 3rd FU: third follow-up.

Table 4

Bivariate and multivariable association between VMDS consumption and fatigue, CoLaus|PsyColaus study, Lausanne, Switzerland.

	Bivariate			Multivariable		
	VMDS Non-users	VMDS Users	P-value	VMDS Non-users	VMDS Users	P-value
2nd FU						
Fatigue_score	2.76 ± 0.04	3.03 ± 0.06	<0.001	2.76 ± 0.04	3.03 ± 0.06	<0.001
Chalder_score	5.53 ± 0.07	6.31 ± 0.11	<0.001	5.58 ± 0.06	6.19 ± 0.11	<0.001
Clinical fatigue (%)	293 (19.3)	141 (25.0)	0.004	1 (ref.)	1.44 (1.13–1.85)	0.004
3rd FU						
Fatigue_score	2.81 ± 0.04	2.95 ± 0.07	0.083	2.81 ± 0.04	2.93 ± 0.07	0.161
Chalder_score	5.51 ± 0.07	5.93 ± 0.12	0.003	5.54 ± 0.07	5.78 ± 0.12	0.088
Clinical fatigue (%)	249 (18.8)	98 (24.3)	0.017	1 (ref.)	1.40 (1.06–1.86)	0.019

Results are expressed as average ± standard deviation or number of participants (column percentage) for bivariate analysis and as multivariable-adjusted average ± standard error or odds ratio and 95 % confidence interval. Between group comparison using ANOVA or t-test for bivariate analyses and ANOVA or logistic regression for multivariable analyses, adjusting for age, sex, BMI categories (normal, overweight, obese), education (low/medium/high), marital status (alone, in couple), smoking (never, former, current), alcohol consumption (none, 1–13, 14–27 and 28+ per week), hypertension (yes, no), diabetes (yes, no), total energy intake (continuous), Alternate Healthy Eating Index (continuous). Abbreviations: VMDS: Vitamin/mineral and/or dietary supplements; 2nd FU: second follow-up; 3rd FU: third follow-up.

Maric et al. showed that multivitamin and mineral supplementation decreased fatigue in patients with chronic fatigue syndrome [28].

However, unlike the above positive results, a randomized controlled trial showed that oral vitamin D3 did not improve markers of vascular health or fatigue in patients with chronic fatigue syndrome [29]. Data from Brouwers et al. showed no statistically significant relationship between symptoms of chronic fatigue syndrome and multivitamins and mineral supplementation³⁰. Also, a meta-analysis showed that vitamin and mineral status did not lead to clinical improvement in chronic fatigue syndrome [31]. Hence, our results confirm the negative findings of the previous studies [29–31], suggesting that VMS is not associated with a lower fatigue level.

4.2. Association between VMDS consumption and fatigue

In addition to the vitamin and mineral supplements mentioned above, some dietary supplements such as dietary polysaccharides, coenzyme Q₁₀, octacosanol, and omega-3 poly-unsaturated fatty acids have also been suggested to reduce the burden of fatigue [32–35]. However, a systematic review found limited evidence that dietary supplements can help relieve symptoms of chronic fatigue syndrome, findings being inconsistent across studies [36]. Another study showed no significant association between dietary supplement intake and fatigue severity or functional impairment [37].

In our study, participants taking VMDS tended to present with higher fatigue scores. Our results thus partially confirm the lack of

beneficial effect of VMDS on fatigue as suggested previously [36,37]. Still, a reverse causation bias cannot be excluded, participants with higher fatigue score consuming more VMDS in the hope of decreasing fatigue levels.

4.3. Strengths and limitations

This study has several strengths. First, the authors studied two separate periods to assess the associations between consumption of vitamins/minerals and dietary supplements with fatigue. Second, the relatively large sample size allowed an adequate statistical power. Third, it was conducted on a population-based sample, not on specific populations such as athletes [19] or people with chronic fatigue syndrome [28]. Hence, the results of our study might be more generalizable than others [19,28].

However, our study also has some limitations. First, its cross-sectional setting does not allow inferring causal relationships; a prospective study will be needed. Second, large variations and inaccurate reporting of VMS composition prevented us from assessing the amount of vitamins and minerals consumed by participants. Finally, when computing the AHEI, it was not possible to establish if participants took multivitamins regularly for five years. Hence, two extreme approaches were applied, one by considering all multivitamin users as taking them regularly for at least five years, and another by excluding the multivitamin component from the AHEI calculation, however, in both approaches, the results were similar. Future studies should attempt to collect more detailed information on vitamin supplements so that exact intakes can be estimated.

5. Conclusion

In this population-based cross-sectional study, the authors found no consistent association between VMS consumption and fatigue. Participants taking VMDS tended to present with higher fatigue scores.

Disclosure

The authors report no conflict of interest.

Data availability statement

The data of CoLaus|PsyCoLaus study used in this article cannot be fully shared as they contain potentially sensitive personal information on participants. According to the Ethics Committee for Research of the Canton of Vaud, sharing these data would be a violation of the Swiss legislation with respect to privacy protection. However, coded individual-level data that do not allow researchers to identify participants are available upon request to researchers who meet the criteria for data sharing of the CoLaus|PsyCoLaus Datacenter (CHUV, Lausanne, Switzerland). Any researcher affiliated to a public or private research institution who complies with the CoLaus|PsyCoLaus standards can submit a research application to research.colaus@chuv.ch or research.psycolaus@chuv.ch. Proposals requiring baseline data only, will be evaluated by the baseline (local) Scientific Committee (SC) of the CoLaus and PsyCoLaus studies. Proposals requiring follow-up data will be evaluated by the follow-up (multicentric) SC of the CoLaus|PsyCoLaus cohort study. Detailed instructions for gaining access to the CoLaus|PsyCoLaus data used in this study are available at www.colaus-psycolaus.ch/professionals/how-to-collaborate/

Funding

The CoLaus study was supported by research grants from GlaxoSmithKline, the Faculty of Biology and Medicine of Lausanne, the Swiss National Science Foundation (grants 33CS00-122661, 33CS30-139468, 33CS30-148401, 33CS30-177535, 3247730_204523 and 320030_220190) and the Swiss Personalized Health Network (grant 2018DRI01). The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Credit author statement

Sisi Xie: formal analysis, investigation, visualization, writing-original draft. Vanessa Kraege and Pedro Marques-Vidal: conceptualization, supervision, validation, writing-review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

No generative AI technology was used to write this manuscript.

Acknowledgements

The authors would like to thank all the people who participated in the recruitment of the participants, data collection and validation, particularly Nicole Bonvin, Yolande Barreau, Mathieu Firmann, François Bastardot, Julien Vaucher, Panagiotis Antiochos, Cédric Gubelmann, Marylène Bay, Benoît Delabays and Adelin Barrier.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clnesp.2024.10.170>.

References

- [1] Nicholson K, Stewart M, Thind A. Examining the symptom of fatigue in primary care: a comparative study using electronic medical records. *J Innovat Health Inf Jan* 21 2015;22(1):235–43. <https://doi.org/10.14236/jhi.v22i1.91>.
- [2] Finsterer J, Mahjoub SZ. Fatigue in healthy and diseased individuals. *Am J Hosp Palliat Care Aug* 2014;31(5):562–75. <https://doi.org/10.1177/1049909113494748>.
- [3] Haß U, Herpich C, Norman K. Anti-inflammatory diets and fatigue. *Nutrients Sep* 30 2019;11(10). <https://doi.org/10.3390/nu11102315>.
- [4] Firth J, Teasdale SB, Allott K, Siskind D, Marx W, Cotter J, et al. The efficacy and safety of nutrient supplements in the treatment of mental disorders: a meta-review of meta-analyses of randomized controlled trials. *World Psychiatr Oct* 2019;18(3):308–24. <https://doi.org/10.1002/wps.20672>.
- [5] Cowan AE, Jun S, Gahche JJ, Toozé JA, Dwyer JT, Eicher-Miller HA, et al. Dietary supplement use differs by socioeconomic and health-related characteristics among U.S. Adults, NHANES 2011–2014. *Nutrients Aug* 17 2018;10(8). <https://doi.org/10.3390/nu10081114>.
- [6] Tardy AL, Pouteau E, Marquez D, Yilmaz C, Scholey A. Vitamins and minerals for energy, fatigue and cognition: a narrative review of the biochemical and clinical evidence. *Nutrients Jan* 16 2020;12(1). <https://doi.org/10.3390/nu12010228>.
- [7] Firmann M, Mayor V, Vidal PM, Bochud M, Pécoud A, Hayoz D, et al. The CoLaus study: a population-based study to investigate the epidemiology and genetic determinants of cardiovascular risk factors and metabolic syndrome. *BMC Cardiovasc Disord Mar* 17 2008;8:6. <https://doi.org/10.1186/1471-2261-8-6>.
- [8] Droz N, Marques-Vidal P. Multivitamins/multiminerals in Switzerland: not as good as it seems. *Nutr J Mar* 24 2014;13:24. <https://doi.org/10.1186/1475-2891-13-24>.
- [9] Krupp LB, LaRocca NG, Muir-Nash J, Steinberg AD. The fatigue severity scale. Application to patients with multiple sclerosis and systemic lupus erythematosus. *Arch Neurol Oct* 1989;46(10):1121–3. <https://doi.org/10.1001/archneur.1989.00520460115022>.
- [10] Valko PO, Bassetti CL, Bloch KE, Held U, Baumann CR. Validation of the fatigue severity scale in a Swiss cohort. *Sleep Nov* 2008;31(11):1601–7. <https://doi.org/10.1093/sleep/31.11.1601>.
- [11] Dittner AJ, Wessely SC, Brown RG. The assessment of fatigue: a practical guide for clinicians and researchers. *J Psychosom Res Feb* 2004;56(2):157–70. [https://doi.org/10.1016/s0022-3999\(03\)00371-4](https://doi.org/10.1016/s0022-3999(03)00371-4).
- [12] Chalder T, Berelowitz G, Pawlikowska T, Watts L, Wessely S, Wright D, et al. Development of a fatigue scale. *J Psychosom Res 1993;37(2):147–53*. [https://doi.org/10.1016/0022-3999\(93\)90081-p](https://doi.org/10.1016/0022-3999(93)90081-p).
- [13] Morris RK, Wearden AJ, Mullis R. Exploring the validity of the Chalder Fatigue scale in chronic fatigue syndrome. *J Psychosom Res Nov* 1998;45(5):411–7. [https://doi.org/10.1016/s0022-3999\(98\)00022-1](https://doi.org/10.1016/s0022-3999(98)00022-1).
- [14] Cella M, Chalder T. Measuring fatigue in clinical and community settings. *J Psychosom Res Jul* 2010;69(1):17–22. <https://doi.org/10.1016/j.jpsychores.2009.10.007>.
- [15] Loge JH, Ekeberg O, Kaasa S. Fatigue in the general Norwegian population: normative data and associations. *J Psychosom Res Jul* 1998;45(1):53–65. [https://doi.org/10.1016/s0022-3999\(97\)00291-2](https://doi.org/10.1016/s0022-3999(97)00291-2).
- [16] De Vries J, Michielsens HJ, Van Heck GL. Assessment of fatigue among working people: a comparison of six questionnaires. *Occup Environ Med Jun* 2003;60(Suppl 1):i10–5. https://doi.org/10.1136/oem.60.suppl_1.i10. Suppl 1.
- [17] Stouten B. Identification of ambiguities in the 1994 chronic fatigue syndrome research case definition and recommendations for resolution. *BMC Health Serv Res May* 13 2005;5:37. <https://doi.org/10.1186/1472-6963-5-37>.
- [18] McCullough ML, Feskanich D, Stampfer MJ, Giovannucci EL, Rimm EB, Hu FB, et al. Diet quality and major chronic disease risk in men and women: moving toward improved dietary guidance. *Am J Clin Nutr Dec* 2002;76(6):1261–71. <https://doi.org/10.1093/ajcn/76.6.1261>.
- [19] Suzuki M, Itokawa Y. Effects of thiamine supplementation on exercise-induced fatigue. *Metab Brain Dis Mar* 1996;11(1):95–106. <https://doi.org/10.1007/bf02080935>.
- [20] Suh SY, Bae WK, Ahn HY, Choi SE, Jung GC, Yeom CH. Intravenous vitamin C administration reduces fatigue in office workers: a double-blind randomized controlled trial. *Nutr J Jan* 20 2012;11:7. <https://doi.org/10.1186/1475-2891-11-7>.
- [21] Favrat B, Balck K, Breymann C, Hedenus M, Keller T, Mezzacasa A, et al. Evaluation of a single dose of ferric carboxymaltose in fatigued, iron-deficient women—PREFER: a randomized, placebo-controlled study. *PLoS One* 2014;9(4):e94217. <https://doi.org/10.1371/journal.pone.0094217>.
- [22] Al-Eisa ES, Alghadir AH, Gabr SA. Correlation between vitamin D levels and muscle fatigue risk factors based on physical activity in healthy older adults. *Clin Interv Aging* 2016;11:513–22. <https://doi.org/10.2147/cia.S102892>.
- [23] Nowak A, Boesch L, Andres E, Battegay E, Hornemann T, Schmid C, et al. Effect of vitamin D3 on self-perceived fatigue: a double-blind randomized placebo-

- controlled trial. *Medicine (Baltim)* Dec 2016;95(52):e5353. <https://doi.org/10.1097/md.0000000000005353>.
- [24] Roy S, Sherman A, Monari-Sparks MJ, Schweiker O, Hunter K. Correction of low vitamin D improves fatigue: effect of correction of low vitamin D in fatigue study (EViDiF study). *N Am J Med Sci Aug* 2014;6(8):396–402. <https://doi.org/10.4103/1947-2714.139291>.
- [25] Takemoto D, Yasutake Y, Tomimori N, Ono Y, Shibata H, Hayashi J. Sesame lignans and vitamin E supplementation improve subjective statuses and anti-oxidative capacity in healthy humans with feelings of daily fatigue. *Global J Health Sci Mar* 25 2015;7(6):1–10. <https://doi.org/10.5539/gjhs.v7n6p1>.
- [26] Lee MC, Hsu YJ, Shen SY, Ho CS, Huang CC. A functional evaluation of anti-fatigue and exercise performance improvement following vitamin B complex supplementation in healthy humans, a randomized double-blind trial. *Int J Med Sci* 2023;20(10):1272–81. <https://doi.org/10.7150/ijms.86738>.
- [27] Haskell CF, Robertson B, Jones E, Forster J, Jones R, Wilde A, et al. Effects of a multi-vitamin/mineral supplement on cognitive function and fatigue during extended multi-tasking. *Hum Psychopharmacol Aug* 2010;25(6):448–61. <https://doi.org/10.1002/hup.1144>.
- [28] Maric D, Brkic S, Tomic S, Novakov Mikic A, Cebovic T, Turkulov V. Multivitamin mineral supplementation in patients with chronic fatigue syndrome. *Med Sci Mon Int Med J Exp Clin Res Jan* 14 2014;20:47–53. <https://doi.org/10.12659/msm.889333>.
- [29] Witham MD, Adams F, McSwiggan S, Kennedy G, Kabir G, Belch JJ, et al. Effect of intermittent vitamin D3 on vascular function and symptoms in chronic fatigue syndrome—a randomised controlled trial. *Nutr Metabol Cardiovasc Dis Mar* 2015;25(3):287–94. <https://doi.org/10.1016/j.numecd.2014.10.007>.
- [30] Brouwers FM, Van Der Werf S, Bleijenberg G, Van Der Zee L, Van Der Meer JW. The effect of a polynutrient supplement on fatigue and physical activity of patients with chronic fatigue syndrome: a double-blind randomized controlled trial. *QJM Oct* 2002;95(10):677–83. <https://doi.org/10.1093/qjmed/95.10.677>.
- [31] Joustra ML, Minovic I, Janssens KAM, Bakker SJL, Rosmalen JGM. Vitamin and mineral status in chronic fatigue syndrome and fibromyalgia syndrome: a systematic review and meta-analysis. *PLoS One* 2017;12(4):e0176631. <https://doi.org/10.1371/journal.pone.0176631>.
- [32] Zhou Y, Chu Z, Luo Y, Yang F, Cao F, Luo F, et al. Dietary polysaccharides exert anti-fatigue functions via the gut-muscle Axis: advances and perspectives. *Foods Aug* 17 2023;(16):12. <https://doi.org/10.3390/foods12163083>.
- [33] Castro-Marrero J, Segundo MJ, Lacasa M, Martinez-Martinez A, Sentañes RS, Alegre-Martin J. Effect of dietary coenzyme Q10 plus nadh supplementation on fatigue perception and health-related quality of life in individuals with myalgic encephalomyelitis/chronic fatigue syndrome: a prospective, randomized, double-blind, placebo-controlled trial. *Nutrients Jul* 30 2021;13(8). <https://doi.org/10.3390/nu13082658>.
- [34] Zhou Y, Cao F, Wu Q, Luo Y, Guo T, Han S, et al. Dietary supplementation of octacosanol improves exercise-induced fatigue and its molecular mechanism. *J Agric Food Chem Jul* 14 2021;69(27):7603–18. <https://doi.org/10.1021/acs.jafc.1c01764>.
- [35] Maes M, Mihaylova I, Leunis JC. In chronic fatigue syndrome, the decreased levels of omega-3 poly-unsaturated fatty acids are related to lowered serum zinc and defects in T cell activation. *Neuroendocrinol Lett Dec* 2005;26(6):745–51.
- [36] Jones K, Probst Y. Role of dietary modification in alleviating chronic fatigue syndrome symptoms: a systematic review. *Aust N Z J Publ Health Aug* 2017;41(4):338–44. <https://doi.org/10.1111/1753-6405.12670>.
- [37] Goedendorp MM, Knoop H, Schippers GM, Bleijenberg G. The lifestyle of patients with chronic fatigue syndrome and the effect on fatigue and functional impairments. *J Hum Nutr Diet Jun* 2009;22(3):226–31. <https://doi.org/10.1111/j.1365-277X.2008.00933.x>.